# Connor Rowland

## GAMEPLAY AND SYSTEMS PROGRAMMER

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### PROFESSIONAL SUMMARY

Gameplay and systems programmer graduating May 2027 (BFA, Digital Game Design and Development, LIU Honors College, GPA 3.9, full-ride scholarship). Lead Unity Developer (intern) on Cubilete Dice Game, a multiplayer mobile game live on iOS and Android, with 8 releases shipped. Founder and lead developer of Perihelion, a Unity rocket-engine simulation shown at NYC Games Festival 2026. Also writes embedded C++ flight-computer firmware.

### WORK EXPERIENCE

**Cubiletecup LLC** June 2026 - Present

**Lead Unity Developer (Intern)** · Remote

- Shipped 8 releases (v0.3 to v2.0.2) of Cubilete Dice Game to iOS and Android on a three-person team; the game is live on both stores.
- Led the v1.0 to v2.0 backend cutover with zero downtime: migrated data to a new Firebase schema, added 16 Cloud Functions, kept v1.0 live for players while v2.0 was in App Store review, and checked about 300 migrated records.
- Built the UI side of the v2.0 rebuild in UI Toolkit from a teammate's blockout: 18 screens, the component foundation, and menu motion and sound, integrating a 142-file final art set.
- Built most of the player-presence system on Firebase Realtime Database and Firestore; coordinate 3-6-player multiplayer tests and fixed the 4 or so bugs they found, including a Photon Fusion bug that split every room of more than 2 players into 2 lobbies.
- Cleared 9 Apple rejections across v0.3-v1.2; on my submissions, first rejection to approval fell from 12 to 6 to 2 days, and v2.0 passed review on the first try.
- Built most of the Sentry error reporting with a teammate; routed every swallowed catch and async-void entry point through one ErrorReporter, finding and fixing 3 silent bugs.
- Finished the Spanish localization (the last 62 of 404 UI strings) and fixed the layout on iPads and tablets.

**Tritium Softworks** May 2025 - Present

**Founder and Lead Developer** · Remote

- Lead a three-person team (physics developer, UI/texturing artist) on Perihelion, a Unity rocket-engine simulation coming to Steam; 176 wishlists and 34 Discord members as of October 2026.
- Built a fluid-network solver that walks propellant flow through pumps, preburners, pipes and tees, reporting 15 failure modes (starved injectors, propellant mixing, nozzle burnthrough) across 6 part types, 4 pipe types and 3 propellant combinations.
- Rewrote the part-attachment system with attach nodes, a parented part tree and type-validated connections; attachment bugs went from about 10 to 0, with no limit on part count.
- Rebuilt the engine plume as a mesh-based Shader Graph effect driven by throttle, mixture and chamber pressure; across that update, 1% low FPS rose 15x (28 to 420) and frame time fell 71% (6.5 to 1.9 ms).
- Built a Unity editor window that generates part icons, with parts packaged and loaded as AssetBundles; create the game's 3D models.
- Showcased at NYC Games Festival (May 2026), LIU Spring Showcase (April 2026) and LIU Winter Playtest (December 2025), about 10 players at each.

### PROJECTS

**Apocamow: Replanted** Solo capstone, Sept 2026 - Present

Unity 6 rebuild of my freshman-year game. Built a GPU grass system on a 512x512 height map that regrows and is cut and trampled; it runs at 95 FPS and 3.4 ms GPU frame time in the editor (RTX 4070 SUPER), with 13,333 instanced grass draws batched into 183 draw calls.

Unity 6, C#, Shader Graph

**Hephaestus Flight Computer** Dec 2025 - Present

Wrote C++ firmware for a BMP280 barometer and two MPU6050 IMUs with a complementary filter and 60 Hz telemetry, SD-card logging, and a vertical-speed estimate that fuses accelerometer integration with barometric altitude.

C++, Arduino (Teensy), I2C

**Game portfolio** crdgd.itch.io

25 published projects with 1,591 views, 119 downloads and 31 followers, including RoboZoo (programmer on a five-person team: 5 attachment types and state management across 3 scenes).

### EDUCATION

**BFA, Digital Game Design and Development** Long Island University, Honors College Sept 2023 - May 2027 (expected)

GPA 3.9/4.0. Full-ride scholarship. Dean's List, Fall 2023 - Fall 2025.

### LEADERSHIP & AWARDS

Head, Showcase Committee, Long Island University (2025 - Present): ran 3 playtest and showcase events, each with about 40 attendees and 32 games.

Organizer, 3x3 Jam (2025 - 2026): a three-week, three-engine game jam with 18 entries.

SkillsUSA New York State Winner (2023), Interactive Application and Video Game Development.

### SKILLS

**Languages:** C#, C++, Python, JavaScript, HTML/CSS

**Unity:** UI Toolkit, Shader Graph, editor scripting, AssetBundles, Unity Profiler, GPU instancing

**Backend and services:** Firebase (Realtime Database, Firestore, Cloud Functions), Sentry, Photon Fusion

**Tools:** Git, Docker, Blender, Maya, Figma, OpenRocket